# Research Proposal

LLM-Quantified Trump Policy Shocks and
Multi-Asset Market Impact Analysis:

*A Three-Paper Series on Equities, Commodities, and Cryptocurrency*

April 2026

## Overview

This document outlines a three-paper research series that employs large language models (LLMs) to quantify the market impact of Trump administration policy communications across three major asset classes. The unifying methodological innovation is a shared LLM-based news analysis pipeline that transforms unstructured news text into a multi-dimensional Trump Policy Shock Index (TPSI), which is then used as the primary explanatory variable in each paper's empirical analysis. The three papers target distinct asset classes—equity indices, commodity futures, and cryptocurrencies—while sharing a common sample period (November 2024 to March 2026), news corpus, and geopolitical lens spanning the United States, mainland China, and Hong Kong.

The 2025–2026 Trump administration has generated an unusually dense sequence of policy events with direct financial market implications: the April 2, 2025 "Liberation Day" tariff announcement, escalating tariffs on Chinese imports reaching 100%, the February 20, 2026 Supreme Court ruling that IEEPA does not authorize tariffs, subsequent Section 232 and Section 122 tariff pivots, and simultaneous pro-cryptocurrency executive orders including the Strategic Bitcoin Reserve initiative. This extraordinary policy environment provides rich variation for identifying the causal channels through which political rhetoric and action transmit to asset prices.

Existing literature predominantly relies on the Caldara and Iacoviello (2022) Geopolitical Risk (GPR) index—a keyword-counting measure derived from ten English-language newspapers—to quantify geopolitical risk in financial markets. While influential, this approach cannot distinguish between policy types (tariffs versus sanctions versus diplomatic rhetoric), capture semantic nuance (threats versus implemented actions), or process non-English sources. Large language models offer a fundamentally different approach: they can parse context, identify causal claims, classify policy types, and score sentiment intensity with chain-of-thought reasoning. This series exploits that capability to construct richer, more granular measures of policy shock.

# Paper 1: One Shock, Three Markets

*LLM-Based Decomposition of Trump Policy Sentiment and Asymmetric Equity Index Responses Across the US, Hong Kong, and Mainland China*

## 1. Introduction

A growing body of work demonstrates that large language models can extract financially relevant sentiment from news text with accuracy rivaling or exceeding human annotators (Lopez-Lira and Tang, 2023; Chen et al., 2025). Simultaneously, the literature on geopolitical risk and equity markets has established that political shocks affect stock returns and volatility through investor sentiment, risk premia, and trade-flow channels (Caldara et al., 2024; IMF Global Financial Stability Report, April 2025). However, these two strands remain largely disconnected: LLM-based sentiment studies focus on single-market settings, while geopolitical risk studies rely on aggregate indices that obscure the heterogeneous channels through which different policy types affect different markets.

This paper bridges the gap by analyzing how the same Trump policy shock—as classified and scored by an LLM pipeline—propagates differently across three institutionally distinct but geopolitically connected equity markets: the United States (S&P 500, Nasdaq 100), Hong Kong (Hang Seng Index, Hang Seng China Enterprises Index, Hang Seng Tech Index), and mainland China (CSI 300). These markets differ fundamentally in investor composition (institutional versus retail dominance), information environment (English-language versus Chinese-language media filtering), regulatory regime (open capital account versus capital controls), and trading hours (creating natural lead-lag structure). A single Trump tariff announcement thus constitutes a quasi-natural experiment in cross-market information transmission.

The paper's central contribution is the Trump Policy Shock Index (TPSI), constructed from LLM-based multi-dimensional classification of Trump-related news. Unlike the binary positive/negative sentiment used in most LLM finance studies (Zhang et al., 2024; Bie et al., 2026), the TPSI provides sub-indices by policy type (tariff, sanction, diplomatic, crypto-regulatory) and target geography (US-domestic, China, Hong Kong, EU), enabling granular identification of which policy channels drive which markets.

## 2. Literature Review

### *2.1 LLMs in Financial Sentiment Analysis*

The application of LLMs to financial text analysis has accelerated rapidly since 2023. Lopez-Lira and Tang (2023) demonstrated that ChatGPT-derived sentiment from news headlines predicts next-day stock returns, outperforming traditional dictionary-based methods. Zhang et al. (2024) extended this to Chinese financial news, finding that LLM performance varies substantially across model architectures, with domain-specific pre-trained models outperforming general-purpose LLMs on Chinese text. Bie et al. (2026) advanced the field by using DeepSeek to extract asset-specific sentiment intensity from Dow Jones Newswires, showing that a one-standard-deviation increase in news magnitude raises next-day abnormal

trading volume by 17% for bonds and 29% for equities. Chen et al. (2025) proposed a Domain Knowledge Chain-of-Thought (DK-CoT) prompting strategy that integrates financial domain knowledge into LLM reasoning, improving sentiment classification accuracy. A comprehensive review by Nazareth and Reddy (2025) synthesized 84 studies on LLM applications in equity markets, identifying prompting strategies, multi-agent frameworks, and fine-tuning as the three dominant methodological approaches.

Despite this progress, the literature remains overwhelmingly focused on firm-level sentiment in single-market settings. No study has used LLMs to decompose policy-level shocks into multiple dimensions and trace their differential propagation across geopolitically linked markets.

### *2.2 Geopolitical Risk and Equity Markets*

Caldara and Iacoviello (2022) introduced the GPR index, which has become the standard measure in the geopolitics-finance nexus. The IMF's April 2025 Global Financial Stability Report documented that major geopolitical risk events reduce equity valuations by 1–2% on impact in advanced economies, with commodity-exporting countries often experiencing positive stock returns due to terms-of-trade effects. Palomba and Tedeschi (2025) extended the analysis to cross-country spillovers using TVP-VAR networks, finding that the Russia-Ukraine conflict triggered a structural break in geopolitical risk transmission. For the US-China dimension specifically, Clayton et al. (2023) used LLMs to classify geoeconomic instruments (tariffs versus export controls) and trace firm-level adjustments, while the BIS Working Paper 1294 (2025) demonstrated that LLMs can decompose macroeconomic sentiment into supply and demand drivers with correlations exceeding 0.8 against established econometric benchmarks.

### *2.3 Cross-Market Information Transmission*

The speed at which information transmits across international equity markets has been studied extensively, but primarily for earnings announcements and macroeconomic data releases (Wongswan, 2006; Bae et al., 2012). For geopolitical shocks, the evidence is thinner. The institutional structure of the US-HK-China triad creates a particularly interesting setting: Hong Kong operates under a linked exchange rate system with the US dollar, serves as the primary offshore channel for mainland Chinese capital, and hosts dual-listed securities (AH shares) that provide direct arbitrage links. Trump's tariff announcements during US trading hours are first reflected in US markets, then in Hong Kong (which opens before mainland China), and finally in the CSI 300—creating a natural laboratory for studying information transmission speed and attenuation.

## 3. Research Questions and Hypotheses

**RQ1:** How do LLM-classified Trump policy shocks (tariff, sanction, diplomatic, crypto-regulatory) differentially affect equity index returns and volatility in the US, Hong Kong, and mainland China?

**RQ2:** What is the speed and magnitude of cross-market information transmission for different types of Trump policy shocks?

**RQ3:** Does the LLM-constructed TPSI provide incremental predictive power over the Caldara-Iacoviello GPR index and the Baker-Bloom-Davis Economic Policy Uncertainty (EPU) index?

***H1a***: *Tariff-related TPSI shocks have the largest negative impact on Hong Kong and mainland Chinese indices, and a smaller but significant negative impact on the Nasdaq 100 (due to supply-chain exposure), while the S&P 500 exhibits a more muted response.*

***H1b***: *Sanction-related TPSI shocks disproportionately affect the HSCEI (mainland enterprises listed in Hong Kong), as these firms face direct exposure to US financial restrictions.*

***H2***: *Information transmission follows the sequence US → HK → China, with Hong Kong absorbing approximately 60–80% of the US market's response within the first trading session and CSI 300 absorbing the remainder with a one-day lag.*

***H3***: *The TPSI sub-indices collectively dominate the GPR index in explaining cross-market return variation, as measured by incremental R-squared and Diebold-Mariano forecast comparison tests.*

## 4. Data

| Variable | Source | Frequency | Period |
|---|---|---|---|
| Trump-related news (EN) | ForexFactory, Yahoo News | Intraday | Nov 2024 – Mar 2026 |
| Trump-related news (CN) | jin10.com | Intraday | Nov 2024 – Mar 2026 |
| S&P 500 (SPX) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| Nasdaq 100 (NDX) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| Hang Seng Index (HSI) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| Hang Seng CEI (HSCEI) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| Hang Seng Tech (HSTECH) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| CSI 300 | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| GPR, GPT, GPA indices | Caldara-Iacoviello website | Daily | Nov 2024 – Mar 2026 |
| EPU index | Baker, Bloom, Davis website | Monthly | Nov 2024 – Mar 2026 |
| Controls: VIX, DXY, US 10Y, CNY/USD | FRED, Bloomberg | Daily | Nov 2024 – Mar 2026 |

*Table 1. Data sources for Paper 1.*

## 5. Methodology

### *5.1 LLM-Based Index Construction*

The TPSI is constructed from a five-stage LLM pipeline applied to the full Trump-related news corpus. Stage 1 uses Claude Haiku 4.5 for binary relevance screening (estimated cost: ~$40 for 10,000 articles). Stages 2–4 use Claude Sonnet 4.6 with structured prompting to classify policy type, tag target geography, and score sentiment direction (−5 to +5) and intensity (low/medium/high). Each classification includes a chain-of-thought reasoning trace to ensure interpretability. Stage 5 aggregates article-level scores into daily TPSI values and sub-indices (TPSI_tariff, TPSI_sanction, TPSI_diplomatic, TPSI_crypto) by computing volume-weighted average sentiment, where volume weights reflect the publishing outlet's reach.

To validate the LLM pipeline, we randomly sample 500 articles for independent human annotation by two coders. We report Cohen's kappa for inter-rater reliability (target: $\kappa > 0.75$) and benchmark the LLM's classifications against the human consensus. As a robustness check, we replicate the full pipeline using DeepSeek V3.2 to assess whether results are model-dependent.

### *5.2 Event Study*

We identify major Trump policy events from the TPSI time series (defined as days where TPSI exceeds two standard deviations from its rolling 30-day mean) and compute cumulative abnormal returns (CARs) over [−5, +10] day windows for each equity index. Abnormal returns are estimated using a market model with the MSCI World Index as the benchmark. We test for cross-market differences in CARs using Kolari and Pynnonen's (2010) adjusted t-test, which accounts for cross-sectional correlation and event-induced volatility.

### *5.3 Panel Regression*

We estimate daily panel models with market-index fixed effects:

$$R_{i,t} = \alpha_i + \beta_1 \, TPSI_tariff_t + \beta_2 \, TPSI_sanction_t + \beta_3 \, TPSI_diplomatic_t + \gamma' X_t + \varepsilon_{i,t}$$

where $R_{i,t}$ is the daily return on index i, $X_t$ is a vector of controls (VIX, DXY, US 10-year yield, CNY/USD), and standard errors are clustered by date. We test for coefficient heterogeneity across markets using triple-interaction terms (TPSI × market × policy type).

### *5.4 Cross-Market Transmission*

To quantify the speed of information transmission, we estimate pairwise Granger causality tests between the CARs of US, HK, and China indices, conditioning on TPSI sub-index regimes (high versus low shock periods). We further estimate the Diebold-Yilmaz (2012) connectedness index to compute time-varying directional spillovers across the three markets.

### *5.5 Benchmark Comparison*

We conduct a formal horse race between the TPSI and the Caldara-Iacoviello GPR index by estimating nested models: (a) GPR-only, (b) TPSI-only, and (c) combined. We compare in-sample fit using incremental R-squared and AIC/BIC, and out-of-sample forecast accuracy (70/30 split) using Diebold-Mariano tests.

## 6. Expected Contributions

- First study to analyze LLM-decomposed policy shocks across three geopolitically connected equity markets (US, HK, China) simultaneously, exploiting the unique institutional structure of each market for identification.
- Construction of a novel multi-dimensional Trump Policy Shock Index (TPSI) with policy-type and geography sub-indices, offering substantially richer information content than binary sentiment or keyword-based GPR measures.
- Empirical evidence on the speed and attenuation of cross-market information transmission for geopolitical policy news, leveraging the natural time-zone sequencing of US → HK → China trading sessions.
- Methodological contribution to the growing LLM-in-finance literature by demonstrating multi-LLM validation (Claude versus DeepSeek) and structured chain-of-thought prompting for policy event classification.

## 7. Target Journals

Primary: Journal of International Money and Finance; Journal of Banking & Finance. Alternative: Pacific-Basin Finance Journal; International Review of Financial Analysis.

# Paper 2: Beyond the GPR Index

*LLM-Constructed Trump Policy Shock Indicators and Heterogeneous Impacts on Precious Metals and Energy Futures*

## 1. Introduction

The relationship between geopolitical risk and commodity prices is among the most extensively studied topics in financial economics. Dozens of papers have employed the Caldara and Iacoviello (2022) GPR index to examine how geopolitical events affect crude oil (Baumeister and Kilian, 2016; Dallas Fed Working Paper 2024), gold (Baur and Smales, 2020; Shahzad et al., 2023), and broader commodity baskets (Wang et al., 2022; Palomba and Tedeschi, 2025). A robust consensus has emerged: geopolitical risk raises commodity price volatility, with gold serving as a safe-haven asset and crude oil responding primarily through supply-disruption channels.

However, this literature faces a fundamental measurement problem. The GPR index counts the frequency of pre-specified keywords (e.g., "war," "threat," "military") across ten English-language newspapers. It cannot distinguish between a tariff announcement and a military escalation, between a policy threat and an implemented action, or between rhetoric directed at China and rhetoric directed at Europe. These distinctions matter enormously for commodity markets: a tariff on Chinese goods affects crude oil through demand-contraction expectations, while a military threat in the Strait of Hormuz affects it through supply-disruption fears. Gold responds to both, but through different mechanisms (safe-haven demand versus inflation hedging).

This paper directly addresses this measurement gap by constructing LLM-based Trump Policy Shock sub-indices and benchmarking them against the GPR index in a formal horse race. The 2025–2026 period offers an ideal testing ground: Trump's tariff escalations (April 2025 Liberation Day, October 2025 100% China tariff threat), the February 2026 IEEPA Supreme Court ruling, and subsequent policy pivots constitute a sequence of well-defined, temporally separated shocks that allow clean identification of the TPSI's incremental information content.

## 2. Literature Review

### *2.1 Geopolitical Risk and Commodity Markets*

The foundational work by Caldara and Iacoviello (2022) established the GPR index as the standard measure of geopolitical risk in financial research. Subsequent studies have applied it extensively to commodity markets. Mitsas et al. (2025) used EGARCH models to show that GPR shocks increase the volatility of gold, silver, and natural gas futures, while wheat, corn, and crude oil exhibit positive return effects. Palomba and Tedeschi (2025) employed TVP-VAR connectedness analysis across 13 commodity futures and found that the Russia-Ukraine conflict triggered a structural break in cross-commodity spillovers, with the BRICS nations and the US emerging as primary spillover transmitters.

Critically, Caldara and Iacoviello (2022) themselves decomposed the GPR into geopolitical threats (GPT) and geopolitical acts (GPA), finding differential effects on financial markets. Caldara et al. (2025) further showed that geopolitical threats drive commodity prices upward through precautionary demand, while geopolitical acts produce more complex effects depending on the specific commodity. However, the GPT/GPA decomposition remains keyword-based and cannot capture the semantic richness of modern policy communications.

### *2.2 LLMs for Commodity Market Analysis*

The application of LLMs specifically to commodity markets is nascent. An arXiv paper by researchers at multiple institutions (2025) proposed an agentic AI framework that fuses news embeddings with price time series for commodity price shock forecasting, finding that removing the news embedding component caused model accuracy to collapse from high performance to near-random (F1 = 0.08). Permutable AI (2025) demonstrated real-time LLM-based analysis of the gold-oil correlation during geopolitical stress periods, showing that the traditional positive correlation broke down during the India-Pakistan tensions of May 2025. The BIS Working Paper 1294 (2025) established a broader precedent by using LLMs to decompose macroeconomic sentiment into supply and demand drivers, achieving correlations above 0.8 with established econometric shock decompositions.

No study has yet constructed LLM-based geopolitical shock indices and formally benchmarked them against the GPR index for commodity market analysis.

## 3. Research Questions and Hypotheses

**RQ1:** Do LLM-constructed TPSI sub-indices (tariff, sanction, diplomatic) exhibit heterogeneous effects on gold, silver, and crude oil futures returns and volatility?

**RQ2:** Does the TPSI outperform the Caldara-Iacoviello GPR index (and its GPT/GPA sub-components) in explaining and forecasting commodity futures dynamics?

**RQ3:** How do commodity markets respond asymmetrically to Trump policy threats versus implemented actions, as classified by LLM?

**RQ4:** What is the cross-commodity spillover pattern of Trump policy shocks?

***H1a****: Tariff-related TPSI shocks negatively affect crude oil returns through demand-contraction expectations, while positively affecting gold returns through safe-haven rebalancing.*

***H1b****: Sanction-related TPSI shocks positively affect both gold and silver through safe-haven and monetary-debasement channels, with silver exhibiting a larger response due to its dual industrial-monetary nature.*

***H2**: Models incorporating TPSI sub-indices achieve statistically superior out-of-sample forecast accuracy compared to GPR-only models, as the LLM captures semantic distinctions that keyword counting misses.*

***H3**: LLM-classified threats generate immediate volatility increases across all three commodities, while implemented actions produce mean-reverting volatility patterns as uncertainty resolves.*

***H4**: Cross-commodity spillovers are stronger during high-TPSI periods, with gold serving as the primary transmitter to silver and crude oil serving as a net receiver of geopolitical shocks.*

## 4. Data

| Variable | Source | Frequency | Period |
|---|---|---|---|
| Trump-related news corpus | ForexFactory, Yahoo News, jin10.com | Intraday | Nov 2024 – Mar 2026 |
| Gold futures (GC) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| Silver futures (SI) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| WTI crude oil futures (CL) | TradingView / Yahoo Finance | Daily OHLCV | Nov 2024 – Mar 2026 |
| GPR, GPT, GPA indices | Caldara-Iacoviello | Daily | Nov 2024 – Mar 2026 |
| Controls: DXY, VIX, US 10Y, Fed Funds Rate, S&P 500 | FRED | Daily | Nov 2024 – Mar 2026 |
| CFTC Commitments of Traders | CFTC | Weekly | Nov 2024 – Mar 2026 |

*Table 2. Data sources for Paper 2.*

## 5. Methodology

### *5.1 TPSI Construction*

The TPSI is constructed using the shared LLM pipeline described in Paper 1. For this paper, we additionally construct TPSI_threat and TPSI_action sub-indices by prompting the LLM to classify each policy event as a "threat" (announced intention without implementation), "action" (implemented policy change), or "rescolution" (withdrawal or judicial reversal of policy). This three-way classification directly competes with the Caldara-Iacoviello GPT/GPA decomposition but leverages the LLM's ability to parse conditional language, policy reversals, and implementation details that keyword counting cannot capture.

### *5.2 EGARCH Modeling*

For each commodity $c \in \{\text{gold, silver, crude oil}\}$, we estimate an EGARCH(1,1) model with TPSI sub-indices as exogenous regressors in both the mean and variance equations:

$$r^c_t = \mu + \Sigma_j \varphi_j \cdot TPSI_{j,t} + \gamma' X_t + \varepsilon_t$$
$$\ln(\sigma^2_t) = \omega + \alpha|z_{t-1}| + \gamma z_{t-1} + \beta \ln(\sigma^2_{t-1}) + \Sigma_j \delta_j \cdot TPSI_{j,t}$$

where j indexes TPSI sub-indices (tariff, sanction, diplomatic, threat, action), $X_t$ is a vector of controls, and the asymmetry parameter $\gamma$ captures leverage effects. The coefficients $\delta_j$ in the variance equation capture how each type of Trump shock affects volatility above and beyond its return impact.

### *5.3 Horse Race: TPSI versus GPR*

We conduct a comprehensive model comparison across three specifications: (a) GPR-only models using the headline GPR index and its GPT/GPA sub-components, (b) TPSI-only models using our LLM-constructed sub-indices, and (c) combined models. In-sample comparison uses incremental R-squared, AIC, and BIC. Out-of-sample comparison uses a rolling 70/30 train-test split with Diebold-Mariano (1995) tests for forecast accuracy and Giacomini-White (2006) tests for conditional predictive ability. We report results for both return forecasting and volatility forecasting.

### *5.4 Cross-Commodity Spillover*

We estimate a TVP-VAR across gold, silver, and crude oil returns, with TPSI sub-indices as exogenous variables, using the Diebold and Yılmaz (2012, 2014) connectedness framework. This produces time-varying total and directional spillover indices that reveal: (a) whether Trump tariff shocks increase cross-commodity connectedness, (b) which commodity serves as the primary transmitter versus receiver, and (c) how the spillover network topology changes during high-TPSI versus low-TPSI periods.

### *5.5 Natural Experiment Analysis*

Three events provide particularly clean identification:

- **April 2, 2025 (Liberation Day):** Broad tariff announcement. TPSI classifies as high-intensity tariff action with China and EU geographic tags. Tests H1a (oil demand contraction) and H1b (gold safe-haven).
- **October 2025 (100% China tariff threat):** TPSI classifies as high-intensity tariff threat with China geographic tag. Tests H3 (threat versus action asymmetry).
- **February 20, 2026 (IEEPA Supreme Court ruling):** TPSI classifies as policy resolution. Tests whether uncertainty resolution reduces commodity volatility as predicted by the precautionary demand channel.

## 6. Expected Contributions

- First formal benchmarking of LLM-constructed policy shock indices against the Caldara-Iacoviello GPR index for commodity market analysis, providing direct evidence on whether semantic understanding outperforms keyword counting in capturing geopolitical risk.
- Novel decomposition of Trump policy shocks into tariff, sanction, and diplomatic channels with commodity-specific heterogeneous effects, revealing that aggregate geopolitical risk indices obscure economically meaningful variation.

- Refined threat-versus-action-versus-resolution classification using LLM chain-of-thought reasoning, advancing the GPT/GPA framework with deeper semantic discrimination and a third category (resolution) that captures uncertainty-reducing events.
- Cross-commodity spillover analysis conditional on Trump policy shock types, revealing transmission channels (e.g., gold → silver safe-haven contagion versus oil → gold inflation hedging) missed by aggregate approaches.

## 7. Target Journals

Primary: Journal of Commodity Markets; Energy Economics. Alternative: Journal of Futures Markets; Resources Policy.

# Paper 3: The Trump Crypto Paradox

*LLM-Quantified Policy Sentiment, the "Trump Trade" Hypothesis, and Tail Risk in Digital Asset Markets*

## 1. Introduction

The Trump administration has maintained an unprecedented dual relationship with cryptocurrency markets. On one hand, pro-crypto executive orders, the Strategic Bitcoin Reserve initiative, appointment of crypto-friendly regulators, and stablecoin legislation have fueled the "Trump trade" narrative (Krause, 2025). On the other hand, aggressive tariff policies have triggered massive crypto sell-offs, including a $19 billion BTC/ETH liquidation following the China tariff escalation in early 2025. This paradox—the same political actor simultaneously driving bullish policy expectations and bearish macro shocks—creates a unique identification challenge that existing studies have not adequately addressed.

Two recent papers have begun to quantify this dynamic. First, a ScienceDirect study (2026) using high-frequency (1-minute) event-study design found that Trump speech sentiment dampens immediate price volatility while simultaneously increasing downside tail-risk exposure—what the authors term the "calm-but-extreme" effect. Second, Krause (2025) documented that Bitcoin's cumulative abnormal returns following Trump events vary significantly by event type: political symbolic gestures (4.49%) generated stronger responses than regulatory relief (2.21%) or executive actions (0.81%). These findings suggest that Bitcoin markets are sensitive to Trump policy communications, but through channels that differ qualitatively from those operating in equity or commodity markets.

This paper extends both studies within the unified TPSI framework. By decomposing Trump news into pro-crypto (regulatory relief, Strategic Bitcoin Reserve, stablecoin legislation) and anti-market (tariffs, sanctions, trade war rhetoric) channels, we can directly test whether the calm-but-extreme effect is channel-specific and whether the Trump trade hypothesis holds after controlling for macro risk. We add a third dimension absent from the existing literature: the geopolitical channel linking US-China tariff escalation to crypto market activity through Chinese capital flight, proxied by the USDT OTC premium. Finally, we extend the analysis beyond Bitcoin and Ethereum to include Monero (XMR), a privacy-focused cryptocurrency whose use case as a sanctions-evasion or capital-flight tool creates a direct theoretical link to Trump's geopolitical policy actions.

## 2. Literature Review

### *2.1 Political Risk and Cryptocurrency Markets*

The relationship between political uncertainty and cryptocurrency returns has evolved rapidly as the crypto market has matured. Demir et al. (2018) established that Bitcoin can serve as a hedge against economic policy uncertainty, while Bouri et al. (2017) found safe-haven properties during specific stress periods. More recently, the evidence has become nuanced. During the Russia-Ukraine conflict, Bitcoin and Ethereum initially surged as safe-haven alternatives but subsequently declined as global risk aversion

intensified, suggesting that crypto's safe-haven status is conditional on the type and magnitude of the geopolitical shock (Shahzad et al., 2023).

Trump-specific studies have focused primarily on his social media activity. Huynh (2021), Krause (2025), and Miroshnichenko (2025) found that Trump's tweets affect Bitcoin returns, volatility, and trading volume, with the effect varying by tweet content and market conditions. The ScienceDirect study (2026) advanced this literature by using high-frequency data and formal sentiment analysis of 47 Trump speeches (August 2024 to February 2025), documenting the calm-but-extreme effect: a one-standard-deviation increase in Trump sentiment reduces per-minute BTC realized volatility by approximately 5% while increasing downside tail-risk exposure. However, this study treated all Trump communications as homogeneous and did not decompose pro-crypto versus anti-market channels.

### *2.2 LLMs in Cryptocurrency Analysis*

LLM applications to crypto markets are growing but remain fragmented. Nguyen et al. (2025) constructed Bitcoin market sentiment indicators using ChatGPT analysis of Twitter/X data, finding significant predictive power for Bitcoin returns even after controlling for established sentiment measures. The CryptoTrade framework by Li et al. (2025) introduced a multi-agent LLM system with fact and subjectivity agents, finding that subjective reasoning is more important in bull markets while factual reasoning dominates in bear markets. The LLM crypto nowcasting study (FinTech, 2025) benchmarked five state-of-the-art LLMs across 12 crypto assets, finding that all models performed poorly on highly volatile assets like BTC and ETH for minute-level forecasting, suggesting that LLMs are better suited for sentiment extraction than direct price prediction.

### *2.3 Privacy Coins and Geopolitical Risk*

Privacy-focused cryptocurrencies such as Monero (XMR) occupy a theoretically distinct position in the geopolitics-crypto nexus. Unlike Bitcoin, whose blockchain is transparent and increasingly subject to regulatory surveillance, Monero's ring signatures and stealth addresses provide transaction-level privacy that makes it a potential tool for sanctions evasion and capital flight. The theoretical prediction is unambiguous: Trump policy actions that increase capital controls, sanctions, or cross-border transaction monitoring should disproportionately increase demand for privacy coins. Empirical evidence, however, is sparse, partly because Monero has been delisted from several major exchanges, reducing data availability. This paper contributes by providing the first systematic analysis of Monero's response to LLM-classified geopolitical policy shocks.

## 3. Research Questions and Hypotheses

**RQ1:** How do LLM-classified pro-crypto Trump policy shocks versus anti-market shocks asymmetrically affect Bitcoin, Ethereum, and Monero returns, volatility, and tail risk?

**RQ2:** Does the "calm-but-extreme" effect persist across different Trump policy types, and does it differ between BTC/ETH and privacy-coin XMR?

**RQ3:** Does the US-China tariff escalation channel drive crypto market activity through Chinese capital flight, as proxied by the USDT OTC premium?

**RQ4:** How do cross-asset spillovers from equities and gold to crypto markets vary with Trump policy shock types?

***H1a***: *Pro-crypto TPSI shocks (Strategic Bitcoin Reserve, regulatory relief) increase BTC and ETH returns and reduce short-term volatility, consistent with the Trump trade hypothesis.*

***H1b***: *Anti-market TPSI shocks (tariffs, sanctions) reduce BTC/ETH returns in the short run but increase Monero returns, as capital flight and sanctions-evasion demand channel into privacy coins.*

***H2***: *The calm-but-extreme effect (reduced volatility, increased tail risk) is driven primarily by pro-crypto TPSI shocks, which create complacency in the short run but uncertainty about policy sustainability in the tail.*

***H3***: *China-targeted TPSI_tariff shocks Granger-cause increases in the USDT OTC premium, which in turn Granger-causes increases in BTC and XMR trading volume, consistent with a capital-flight transmission channel.*

***H4***: *During high anti-market TPSI periods, BTC-equity correlation increases (risk-off behavior), while during high pro-crypto TPSI periods, BTC-equity correlation decreases (decoupling), consistent with regime-dependent safe-haven properties.*

## 4. Data

| Variable | Source | Frequency | Period |
|---|---|---|---|
| Trump-related news corpus | ForexFactory, Yahoo News, jin10.com | Intraday | Nov 2024 – Mar 2026 |
| Bitcoin (BTC/USD) | TradingView / Binance | Daily + Hourly | Nov 2024 – Mar 2026 |
| Ethereum (ETH/USD) | TradingView / Binance | Daily + Hourly | Nov 2024 – Mar 2026 |
| Monero (XMR/USD) | Yahoo Finance / Kraken | Daily | Nov 2024 – Mar 2026 |
| USDT OTC premium | OKX / Binance P2P | Daily | Nov 2024 – Mar 2026 |
| Bitcoin spot ETF flows | Farside Investors | Daily | Nov 2024 – Mar 2026 |
| Controls: S&P 500, Gold, DXY, VIX, US 10Y | TradingView / FRED | Daily | Nov 2024 – Mar 2026 |

*Table 3. Data sources for Paper 3.*

## 5. Methodology

### *5.1 Dual-Channel TPSI Construction*

From the shared LLM pipeline, we construct two daily indices: TPSI_pro_crypto captures pro-crypto policy sentiment (Strategic Bitcoin Reserve announcements, SEC enforcement relief, stablecoin legislation, crypto-friendly appointments) and TPSI_anti_market captures trade-war and sanctions rhetoric (tariff escalations, export controls, diplomatic confrontations targeting China/EU). This dual-channel design directly tests whether Bitcoin is a "Trump trade" (responsive to TPSI_pro_crypto) or a traditional risk asset (responsive to TPSI_anti_market).

### *5.2 Asymmetric Volatility and Tail Risk*

We estimate GJR-GARCH(1,1) models for BTC, ETH, and XMR with TPSI_pro_crypto and TPSI_anti_market as exogenous variables in both the mean and variance equations. The GJR specification captures the asymmetry between positive and negative shock impacts on volatility. For tail risk analysis, we estimate Conditional AutoRegressive Value at Risk (CAViaR) models at the 5th and 1st percentiles, following Engle and Manganelli (2004), with TPSI sub-indices as explanatory variables. This directly tests the calm-but-extreme effect: if TPSI_pro_crypto reduces the GARCH variance but increases the CAViaR tail risk, we have evidence that pro-crypto policy creates short-term complacency with embedded downside vulnerability.

### *5.3 China Capital Flight Channel*

We test the capital-flight transmission mechanism in two stages. First, we estimate whether China-specific TPSI_tariff shocks Granger-cause movements in the USDT OTC premium—the spread between the over-the-counter CNY/USDT rate and the official CNY/USD exchange rate, which serves as a proxy for capital-control circumvention activity. Second, we estimate whether USDT premium movements predict subsequent BTC and XMR trading volume on exchanges with significant Chinese user bases. We use local projection impulse response functions (Jordà, 2005) to trace the dynamic response at 1-day, 3-day, and 7-day horizons.

### *5.4 Cross-Asset Dynamic Correlation*

We estimate a DCC-GARCH model (Engle, 2002) across BTC, S&P 500, gold, and DXY to examine how pairwise correlations vary with Trump policy shock regimes. We split the sample into high-TPSI_pro_crypto versus high-TPSI_anti_market periods and test whether BTC-equity correlation is regime-dependent. This addresses a central question in the crypto literature: under what geopolitical conditions does Bitcoin function as a diversifier versus a correlated risk asset?

### *5.5 High-Frequency Event Analysis*

For the five largest TPSI shocks in the sample period, we conduct intraday event studies using hourly BTC and ETH data from Binance. We measure cumulative abnormal returns at 1-hour, 4-hour, 24-hour, and 72-hour horizons, benchmarked against a rolling 30-day hourly return model. We compare crypto reaction speed with equity market (S&P 500 futures) reaction speed to the same Trump news events,

exploiting crypto's 24/7 trading availability as an advantage for capturing immediate market responses that occur outside traditional equity trading hours.

## 6. Expected Contributions

- First unified LLM-based framework to simultaneously test the "Trump trade" hypothesis (pro-crypto policy channel) and the risk-off hypothesis (tariff/sanction channel) in cryptocurrency markets, resolving the identification challenge created by the Trump administration's paradoxical dual relationship with crypto.
- Extension and decomposition of the "calm-but-extreme" effect (ScienceDirect, 2026) by LLM-classified policy type, providing mechanistic understanding of why political sentiment reduces short-term crypto volatility while increasing tail risk.
- First systematic evidence on the China capital flight channel linking US-China tariff escalation to crypto market microstructure via the USDT OTC premium, with Monero's privacy features providing a theoretically motivated additional test.
- Regime-dependent cross-asset correlation analysis revealing when Bitcoin functions as a diversifier versus a correlated risk asset, conditional on the type of Trump policy shock, with practical implications for portfolio construction.

## 7. Target Journals

Primary: Journal of Financial Economics; Finance Research Letters. Alternative: International Review of Financial Analysis; Research in International Business and Finance.

## Cross-Paper Synthesis

### Unified Framework

The three papers share a common theoretical premise: that large language models can extract richer, more granular information from policy news text than existing keyword-based measures, and that this granularity reveals economically meaningful heterogeneity in how different types of policy shocks transmit to different asset classes. The shared LLM pipeline ensures methodological consistency across papers while allowing each paper to make a distinct empirical contribution.

### Comparison Matrix

| Dimension | Paper 1: Equities | Paper 2: Commodities | Paper 3: Crypto |
|---|---|---|---|
| Core question | How do Trump shocks propagate across US/HK/CN equity markets? | Can LLM-TPSI outperform GPR index for commodity analysis? | Is crypto a Trump trade or a risk-off asset? |
| Dependent variables | SPX, NDX, HSI, HSCEI, HSTECH, CSI 300 | Gold (GC), Silver (SI), WTI Crude (CL) | BTC, ETH, XMR |
| Key TPSI sub-indices | By policy type × geography | Tariff/Sanction/Diplomatic + Threat/Action/Resolution | Pro-crypto vs. Anti-market |
| Primary methods | Event study, panel regression, Granger causality | EGARCH, TVP-VAR, Diebold-Mariano tests | GJR-GARCH, CAViaR, DCC-GARCH |
| Unique innovation | Cross-market transmission speed (US→HK→CN) | Horse race: LLM vs. GPR index | Calm-but-extreme decomposition + China capital flight |
| Data frequency | Daily | Daily | Daily + Hourly |
| Sample period | November 2024 – March 2026 | November 2024 – March 2026 | November 2024 – March 2026 |

*Table 4. Cross-paper comparison.*

### Execution Sequence

Paper 2 (Commodities) should be completed first. The rationale is threefold: (a) commodity futures data is the cleanest of the three asset classes, with no microstructure noise from fragmented exchange venues; (b) the GPR horse race provides the clearest and most publishable narrative, as the Caldara-Iacoviello index is the dominant measure in the field; and (c) the methodology section describing the shared LLM pipeline and TPSI construction will serve as the reference for the subsequent two papers, which can cite Paper 2 for methodological details.

Paper 1 (Equities) follows, adding the cross-market dimension. Paper 3 (Crypto) is completed last, as it requires the most novel data sources (USDT OTC premium, Monero) and makes the most ambitious

theoretical claims (capital flight channel, calm-but-extreme decomposition), which benefit from the methodological credibility established by the first two publications.

## Projected Timeline

| Phase | Period | Deliverables |
|---|---|---|
| Infrastructure | April – May 2026 | LLM pipeline, news corpus, human annotation (500 articles), data archival scripts |
| Paper 2 draft | June – August 2026 | TPSI construction, EGARCH + horse race, first submission |
| Paper 1 draft | September – November 2026 | Cross-market analysis, panel regression, first submission |
| Paper 3 draft | December 2026 – February 2027 | Tail risk, capital flight channel, first submission |
| Revisions | March – June 2027 | Respond to referee reports, update data |

*Table 5. Projected execution timeline.*

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